CROSS-MODAL QUERY FUSION AND ALIGNMENT: ADDRESSING MODALITY IMBALANCE THROUGH QUERY-GUIDED FUSION

Anonymous authors
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ABSTRACT

Modality imbalance remains a fundamental challenge in multimodal learning, where dominant modalities disproportionately shape optimization while weaker modalities are underutilized. Existing studies mainly analyze this problem through optimization behavior, training losses, and final task performance, paying comparatively little attention to the intrinsic structure of multimodal representations. In this work, we revisit modality imbalance from a structural perspective and identify two coupled failure modes: representation-level imbalance, in which modalities occupy mismatched low-dimensional subspaces, and fusion-level imbalance, in which shared fusion layers suppress weak-modality contributions through destructive cross-modal entanglement. This decomposition suggests that correcting optimization imbalance alone is insufficient, as improved representation alignment does not guarantee balanced utilization after fusion. To address these two sources of imbalance, we propose Cross-Modal Query Fusion and Alignment (CQFA), a query-guided framework that explicitly separates modality-specific representation learning from controlled cross-modal interaction through staged training and query-partitioned fusion. Specifically, staged objectives over modality-specific expert queries stabilize modality-specific representations and strengthen cross-modal correspondence before full fusion, while dual query-guided fusion streams together with logit-level aggregation alleviate fusion-level imbalance by reducing premature entanglement and hard shared bottlenecks. On CREMA-D and Kinetics-400, CQFA improves upon state-of-the-art results, achieving 89.94/89.94 and 71.85/71.66 ACC/F1, respectively.

1 INTRODUCTION

Multimodal learning aims to jointly model and integrate information from heterogeneous sources such as images, audio, text, and video in order to build richer and more robust representations Baltrušaitis et al. (2019); Caffagni et al. (2024). Multimodal systems are equipped to reflect the complexity of real-world data by capturing complementary cues across modalities Yin et al. (2024). In recent years, this paradigm has gained significant traction, fueled by the availability of large-scale paired datasets and the success of foundation models like CLIP Radford et al. (2021) and CLAP Elizalde et al. (2023). These advancements have yielded significant performance gains across a range of tasks, including visual question answering Ding et al. (2022), cross-modal retrieval Wang et al. (2024), audio-visual processing Du et al. (2025); Wu et al. (2025), and sentiment analysis Das & Singh (2023). Despite these advances, effective multimodal integration remains challenging. Multimodal fusion may still exhibit bias or redundancy, and in some cases multimodal models have been observed to perform worse than strong unimodal baselines, revealing a persistent gap between the promise of multimodal learning and its realized potential Xu et al. (2025).

To mitigate this issue, existing methods address modality imbalance through interventions at different stages, including data-level reweighting or modality selection Wei et al. (2024), feedforward modulation and adaptive fusion Zhang et al. (2024a); Wei et al. (2025b); Zhou et al. (2023), and balancing strategies at the objective or optimization level Wei & Hu (2024); Yang et al. (2024). Despite their differences, these approaches largely share a common assumption: modality imbalance is primarily a training-dynamics problem that can be alleviated by reweighting samples, suppressing dominant

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signals, or reshaping gradients and losses. Although these treatments have yielded empirical improvements, they generally do not disentangle whether performance degradation stems from imbalanced representational structure prior to fusion or from asymmetric modality utilization within the fusion module itself. As a result, existing methods often mitigate the symptoms of imbalance without fully explaining where the imbalance emerges and why weaker modalities can remain underexploited even after the optimization is rebalanced.

This distinction implies that addressing a single level, e.g., improving representation alignment alone, may not resolve downstream modality underutilization after fusion. This motivates separating representation stabilization from fusion control in both training and architecture. To address these coupled sources of imbalance, we propose a framework that explicitly separates representation stabilization from controlled multimodal fusion. To mitigate representation-level imbalance, CQFA introduces a multi-stage training scheme over modality-specific expert queries, combining contrastive alignment, cross-modal matching, and task supervision to stabilize modality-specific representations and strengthen semantic correspondence before full multimodal fusion. To mitigate fusion-level imbalance, CQFA employs a shared Q-Former with distinct learnable expert and fusion queries, so that modality-specific evidence extraction and cross-modal interaction are handled by different query groups. The final prediction is produced through a decoupled logits-fusion strategy rather than by compressing both modalities into a single shared fused feature, thereby preserving modality-specific decision channels and alleviating weak-modality underutilization in the fusion head. Through this joint design of staged optimization and query-based interaction, CQFA addresses both representation-level imbalance and fusion-level imbalance within a unified framework, extending prior work that treats alignment and fusion separately.

To support reproducibility, we will publicly release the implementation, pretrained checkpoints, and evaluation code upon acceptance. Building on this structural view, the main contributions of this study are summarized as follows:

(1) We formulate modality imbalance from a two-level structural perspective, distinguishing representation-level imbalance, reflected by dimension-collapse-induced subspace misalignment and its associated modality gap, from fusion-level imbalance, reflected by weak-modality underutilization caused by collision-induced interference in shared fusion layers.

(2) We develop a geometric analysis framework for diagnosing modality imbalance, combining effective-dimension analysis with principal-angle-based modality-gap measurement to quantify low-dimensional subspace structure, cross-modal misalignment, and their layer-wise evolution.

(3) We propose Cross-Modal Query Fusion and Alignment (CQFA), a query-guided framework that combines learnable expert and fusion queries, multi-stage alignment objectives, and decoupled logits fusion to stabilize modality-specific representations and control cross-modal interaction.

(4) Extensive experiments and representation analyses on CREMA-D, AVE, and Kinetics-400 show that CQFA delivers strong multimodal classification performance and that its gains are associated with improved high-level alignment and post-fusion consistency, rather than simply minimizing the raw backbone modality gap.

2 Related Work

2.1 Modality Imbalance

Recent survey work Xu et al. (2025) suggests that multimodal imbalance methods can be organized according to the stage at which they intervene. Data-level methods operate on the training distribution or batch composition to strengthen weak modalities before feature learning and fusion Wei et al. (2024); Ma et al. (2025). Feed-forward and fusion-level methods modify modality processing or cross-modal interaction during forward propagation, often by adapting unimodal behavior or estimating modality reliability Zhang et al. (2024a; 2023); Fan et al. (2024). Objective-based methods redesign losses to align modalities, improve discriminability, or balance multimodal and unimodal learning signals Yang et al. (2024); Wei & Hu (2024); Yu et al. (2026); Xu et al. (2023); Jiang et al. (2025). Optimization-based methods act directly on update dynamics by reweighting or disentangling gradients to reduce modality competition and cross-modal interference Peng et al. (2022); Li et al. (2023); Fan et al. (2023); Wei et al. (2025a). These studies show that imbalance can be mitigated at

multiple stages, but most taxonomies mainly describe *where* the intervention is applied. They provide limited insight into whether the failure originates from representation-level structural mismatch or from imbalance introduced during multimodal fusion, motivating a more mechanism-oriented formulation.

2.2 Modality Gap

An emerging body of work has examined multimodal imbalance through the geometry of shared representation spaces. Liang et al. Liang et al. (2022) first identified the *modality gap* in multimodal contrastive learning, formalizing it as a systematic discrepancy between the centers of normalized embeddings from different modalities within a common embedding space. Subsequent studies refined this geometric account. In particular, Yi et al. Yi et al. (2025) showed that the modality gap can be more precisely interpreted through subspace geometry and pairwise alignment, demonstrating that under subspace constraints the observed center discrepancy is governed by the smallest principal angle between modality-specific subspaces rather than collapsing to zero. Complementary work has further investigated why such mismatch persists. Jing et al. Jing et al. (2022) introduced *dimensional collapse*, showing that contrastive representations may occupy only a low-dimensional subspace of the ambient embedding space. Building on this perspective, Zhang et al. Zhang et al. (2024b) related modality gap in multimodal models to dimensional collapse and quantified the extent of this collapse through the *effective dimension* derived from the singular spectrum of the representation covariance matrix. Beyond pre-fusion representation geometry, Chaudhuri et al. Chaudhuri et al. (2025) showed that modality collapse can also arise during fusion through cross-modal interference in shared fusion heads, indicating that geometric alignment alone does not guarantee balanced modality utilization.

3 Methodology

3.1 A Two-Level View of Multimodal Imbalance

We explain modality imbalance through two coupled levels: *representation-level imbalance*, where different modalities occupy mismatched feature subspaces, and *fusion-level imbalance*, where the fusion head underuses one modality due to cross-modal interference.

Representation-Level Imbalance. In a standard multimodal framework, disparate modalities are mapped into a shared ambient embedding space $\mathbb{R}^d$. However, due to differences in the intrinsic complexity of heterogeneous signals, representations frequently suffer from *dimension collapse* Zhang et al. (2024b). Let $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_d$ denote the singular values of a modality's representation covariance matrix. The effective dimension is defined mathematically as Zhang et al. (2024b):

$$d_e = \arg\min_{d'} \frac{\sum_{i=1}^{d'} \sigma_i}{\sum_{i=1}^{d} \sigma_i} \geq \gamma, \tag{1}$$

where γ is a variance-explained threshold (e.g., 0.95). Dimension collapse occurs when $d_e \ll d$. Consequently, features from different modalities intrinsically occupy low-dimensional subspaces, effectively forcing $\boldsymbol{z}_x \in \mathcal{S}_x$ and $\boldsymbol{z}_y \in \mathcal{S}_y$, where $\mathcal{S}_x, \mathcal{S}_y \subseteq \mathbb{R}^d$ and $\dim(\mathcal{S}_x), \dim(\mathcal{S}_y) \ll d$. Representation-level imbalance inherently emerges when modalities collapse into mismatched subspaces, i.e., $\mathcal{S}_x \neq \mathcal{S}_y$.

Following Yi et al. (2025), under this subspace constraint, the *modality gap* can be theoretically interpreted as the angular discrepancy between these modality-specific hyperplanes. Specifically, if $\boldsymbol{U}_x$ and $\boldsymbol{U}_y$ act as orthonormal bases of $\mathcal{S}_x$ and $\mathcal{S}_y$, the gap is structurally related to their smallest principal angle:

$$\theta_{\min} = \arccos\big(\sigma_{\max}(\boldsymbol{U}_x^\top \boldsymbol{U}_y)\big). \tag{2}$$

This conceptually establishes representation-level imbalance as a combination of low effective dimensionality and cross-modal subspace misalignment. We formally operationalize this principal-angle view into a computable metric later in Section 3.4.

Fusion-Level Imbalance. Geometric alignment between modalities does not necessarily ensure their balanced utilization during fusion. Following Chaudhuri et al. (2025), we interpret fusion-level imbalance through the lens of cross-modal polysemantic interference in a shared, capacity-limited

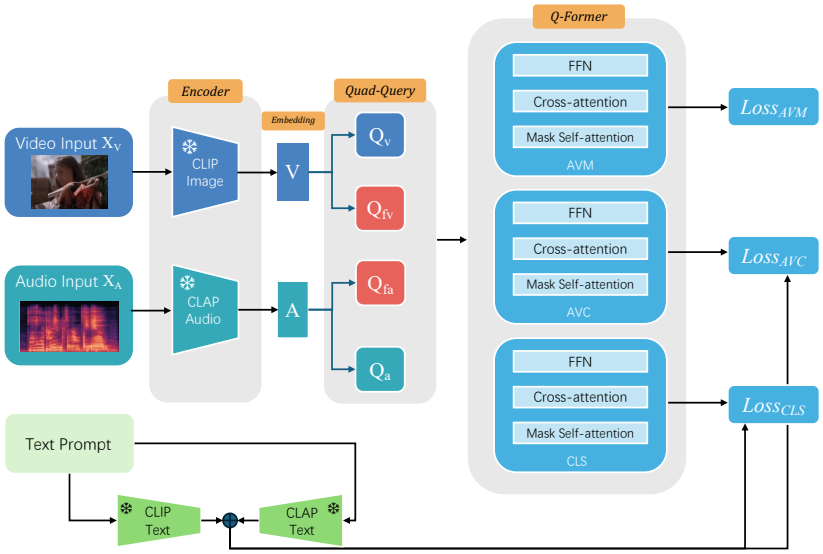


Figure 1: Overview of the CQFA framework.

fusion head. In particular, their Lemma 1 suggests that when fusion neurons are shared across modalities under finite capacity, some neurons can simultaneously encode features from different modalities, giving rise to cross-modal polysemantic collisions. Building on this view, their Theorem 1 further implies that as the proportion of such collisions increases, the marginal contribution of predictive features from a modality to the optimization of the task objective can vanish:

$$\lim_{p(w_p)\to 1} \sum_{\forall \boldsymbol{z}_y \in \mathcal{X}} \frac{\partial}{\partial w_p} \mathcal{L}(\phi(\boldsymbol{z}_y), y) = 0, \tag{3}$$

where w_p denotes polysemantic fusion subspaces or neurons, $p(w_p)$ is the proportion of cross-modal polysemantic collisions, $\boldsymbol{z}_y$ represents predictive features from a modality, and $\mathcal{L}(\phi(\boldsymbol{z}_y), y)$ is the post-fusion task objective. This indicates that collision-induced interference in shared fusion neurons can suppress the optimization contribution of modality-specific predictive signals. We therefore define fusion-level imbalance as the progressive underutilization of a modality within the fusion head caused by such cross-modal interference.

3.2 Framework Overview and Preliminaries

Given a synchronized audio-visual sample $(\boldsymbol{x}_V, \boldsymbol{x}_A)$ with label $y \in \{1, \ldots, C\}$, the visual and audio inputs are encoded by LoRA-tuned modality-specific backbones and mapped by projection heads to obtain modality tokens $\boldsymbol{V} \in \mathbb{R}^{B\times S_V \times D}$ and $\boldsymbol{A} \in \mathbb{R}^{B\times S_A \times D}$. The class-text prototypes $\boldsymbol{g}_t^{\text{clip}}, \boldsymbol{g}_t^{\text{clap}} \in \mathbb{R}^{C\times D}$ are obtained analogously from the corresponding CLIP and CLAP text encoders followed by projection and L2 normalization. These text prototypes provide class-level supervision, but they are not fused with $\boldsymbol{V}$ and $\boldsymbol{A}$ as a third modality.

As shown in Figure 1, CQFA uses a shared Q-Former with four query groups. At layer l, the query state is

$$\boldsymbol{Q}^{(l)} = \left[\boldsymbol{Q}_v^{(l)}; \boldsymbol{Q}_{f,v}^{(l)}; \boldsymbol{Q}_{f,a}^{(l)}; \boldsymbol{Q}_a^{(l)}\right], \qquad \boldsymbol{Q}^{(0)} \text{ is learnable.} \tag{4}$$

The expert queries $\boldsymbol{Q}_v$ and $\boldsymbol{Q}_a$ extract modality-specific evidence from $\boldsymbol{V}$ and $\boldsymbol{A}$, respectively. To make the fusion-query generation explicit, let $S(\boldsymbol{Q})$ denote the Q-Former self-attention update over query sequence $\boldsymbol{Q}$, and let C_v and C_a denote two modality-specific cross-attention transformers:

$$C_v(\boldsymbol{Q}, \boldsymbol{V}) = \text{MHA}_v(\boldsymbol{Q}, \boldsymbol{V}, \boldsymbol{V}), \qquad C_a(\boldsymbol{Q}, \boldsymbol{A}) = \text{MHA}_a(\boldsymbol{Q}, \boldsymbol{A}, \boldsymbol{A}). \tag{5}$$

Here $\boldsymbol{V}$ and $\boldsymbol{A}$ serve as the key-value tokens of the visual and audio cross-attention transformers, respectively, and each updated query stream is further processed by the standard FFN in the Q-Former

layer. After the self-attention update $S(\boldsymbol{Q}^{(l)})$, let $[\widetilde{\boldsymbol{Q}}_v^{(l)}; \widetilde{\boldsymbol{Q}}_{f,v}^{(l)}; \widetilde{\boldsymbol{Q}}_{f,a}^{(l)}; \widetilde{\boldsymbol{Q}}_a^{(l)}]$ denote the resulting query sequence. The two fusion streams are updated as

$$\boldsymbol{Q}_{f,v}^{(l+1)} = C_a\Big(C_v\Big(\widetilde{\boldsymbol{Q}}_{f,v}^{(l)}, \boldsymbol{V}\Big), \boldsymbol{A}\Big), \qquad \boldsymbol{Q}_{f,a}^{(l+1)} = C_v\Big(C_a\Big(\widetilde{\boldsymbol{Q}}_{f,a}^{(l)}, \boldsymbol{A}\Big), \boldsymbol{V}\Big). \tag{6}$$

After the final Q-Former layer, we denote the two fusion-query readouts as $\boldsymbol{q}_{fv} = \boldsymbol{Q}_{f,v}^{(L)}$ and $\boldsymbol{q}_{fa} = \boldsymbol{Q}_{f,a}^{(L)}$.

3.3 Three-Stage Training

CQFA optimizes the shared Q-Former through three stages. AVC trains the expert queries under isolated query interactions to stabilize modality-specific representations. AVM opens the interaction between the two expert-query groups to learn pairwise audio-visual correspondence. CLS activates the two fusion-query groups and routes cross-modal exchange through them for final classification.

As summarized in Appendix Figure 3, the stages differ in active query groups, self-attention masks, and supervision. We reuse the self-attention and cross-attention operators defined in Section 3.2; the standard FFN update is applied inside each Q-Former layer and omitted from the equations for compactness. In stage $s \in \{\mathrm{AVC}, \mathrm{AVM}, \mathrm{CLS}\}$, $\widetilde{\boldsymbol{Q}}_s^{(l)}$ denotes the output of the corresponding stage-specific masked self-attention. The exact mask patterns are specified in Appendix Figure 3.

Stage 1: Audio-Vision Contrastive Learning (AVC). AVC uses the isolated expert-query groups. After masked self-attention, the expert queries cross-attend only to their own modality:

$$\boldsymbol{Q}_v^{(l+1)} = C_v(\widetilde{\boldsymbol{Q}}_v^{(l)}, \boldsymbol{V}), \qquad \boldsymbol{Q}_a^{(l+1)} = C_a(\widetilde{\boldsymbol{Q}}_a^{(l)}, \boldsymbol{A}). \tag{7}$$

The final expert-query outputs $\boldsymbol{Q}_v^{(L)}$ and $\boldsymbol{Q}_a^{(L)}$ are projected into the two text-aligned spaces and L2-normalized to obtain $\boldsymbol{g}_v$ and $\boldsymbol{g}_a$, respectively. Using a learnable temperature τ, the class logits are obtained by query-wise max-pooling over the query-to-text similarities:

$$\boldsymbol{s}_{vt} = \mathrm{MaxPool}_n\Big(\tau\, \boldsymbol{g}_v(\boldsymbol{g}_t^{\mathrm{clip}})^\top\Big), \qquad \boldsymbol{s}_{at} = \mathrm{MaxPool}_n\Big(\tau\, \boldsymbol{g}_a(\boldsymbol{g}_t^{\mathrm{clap}})^\top\Big). \tag{8}$$

Here $\mathrm{MaxPool}_n$ pools over the query dimension and retains, for each class, the largest similarity score among all query tokens, so $\boldsymbol{s}_{vt}, \boldsymbol{s}_{at} \in \mathbb{R}^{B\times C}$. We then apply cross-entropy losses $\mathcal{L}_{vt} = \mathrm{CE}(\boldsymbol{s}_{vt}, y)$ and $\mathcal{L}_{at} = \mathrm{CE}(\boldsymbol{s}_{at}, y)$. To additionally align paired visual and audio query representations, we mean-pool and L2-normalize $\boldsymbol{g}_v$ and $\boldsymbol{g}_a$ to obtain $\bar{\boldsymbol{g}}_v$ and $\bar{\boldsymbol{g}}_a$, and then apply a symmetric contrastive loss:

$$\mathcal{L}_{va} = -\frac{1}{2B}\sum_{i=1}^{B}\left[\log\frac{\exp(\bar{\boldsymbol{g}}_v^{(i)}\cdot\bar{\boldsymbol{g}}_a^{(i)}/\tau)}{\sum_{j=1}^{B}\exp(\bar{\boldsymbol{g}}_v^{(i)}\cdot\bar{\boldsymbol{g}}_a^{(j)}/\tau)} + \log\frac{\exp(\bar{\boldsymbol{g}}_a^{(i)}\cdot\bar{\boldsymbol{g}}_v^{(i)}/\tau)}{\sum_{j=1}^{B}\exp(\bar{\boldsymbol{g}}_a^{(i)}\cdot\bar{\boldsymbol{g}}_v^{(j)}/\tau)}\right]. \tag{9}$$

The AVC objective is

$$\mathcal{L}_{\mathrm{AVC}} = \mathcal{L}_{vt} + \mathcal{L}_{at} + \mathcal{L}_{va}. \tag{10}$$

Stage 2: Audio-Vision Matching (AVM). AVM removes the fusion-query groups and processes only $[\boldsymbol{Q}_v, \boldsymbol{Q}_a]$ with an open interaction mask. This allows the expert queries to exchange information while still cross-attending to their own modalities. The resulting query sequence is fed to a matching head:

$$\boldsymbol{z}_{\mathrm{AVM}} = \mathrm{MLP}_{\mathrm{AVM}}\Big([\boldsymbol{Q}_v^{(L)}; \boldsymbol{Q}_a^{(L)}]\Big) \in \mathbb{R}^{B\times 2}. \tag{11}$$

Given binary labels y_{match} over positive and hard-negative audio-visual pairs, the AVM loss is

$$\mathcal{L}_{\mathrm{AVM}} = \mathrm{CE}(\boldsymbol{z}_{\mathrm{AVM}}, y_{\mathrm{match}}). \tag{12}$$

$\mathrm{CE}(\cdot,\cdot)$ denotes the cross-entropy loss.

Stage 3: Fusion Classification (CLS). CLS activates all four query groups and uses a bottleneck mask that blocks direct $\boldsymbol{Q}_v \leftrightarrow \boldsymbol{Q}_a$ communication. Cross-modal exchange is routed through the dual fusion streams in Equation 6. The final fusion-query outputs $\boldsymbol{q}_{fv}$ and $\boldsymbol{q}_{fa}$ are projected into CLIP- and CLAP-aligned spaces and L2-normalized to obtain $\boldsymbol{g}_{f,v}$ and $\boldsymbol{g}_{f,a}$, respectively. The two decision channels follow the same query-wise pooling rule:

$$\boldsymbol{s}_v = \mathrm{MaxPool}_n\Big(\tau\, \boldsymbol{g}_{f,v}(\boldsymbol{g}_t^{\mathrm{clip}})^\top\Big), \qquad \boldsymbol{s}_a = \mathrm{MaxPool}_n\Big(\tau\, \boldsymbol{g}_{f,a}(\boldsymbol{g}_t^{\mathrm{clap}})^\top\Big). \tag{13}$$

The final prediction uses logits fusion rather than feature concatenation:

$$\boldsymbol{s}_{\text{cls}} = \boldsymbol{s}_v + \boldsymbol{s}_a \in \mathbb{R}^{B\times C}, \qquad \mathcal{L}_{\text{CLS}} = \text{CE}(\boldsymbol{s}_{\text{cls}}, y). \tag{14}$$

This preserves modality-specific decision channels while aggregating evidence at the score level. The total training objective is

$$\mathcal{L}_{\text{total}} = \alpha_{\text{AVC}}\mathcal{L}_{\text{AVC}} + \alpha_{\text{AVM}}\mathcal{L}_{\text{AVM}} + \alpha_{\text{CLS}}\mathcal{L}_{\text{CLS}}. \tag{15}$$

Inference. At inference time, CQFA uses the CLS-stage dual fusion streams and predicts

$$\hat{y} = \underset{c\in\{1,\dots,C\}}{\arg\max}\ \boldsymbol{s}_{\text{cls}}(c). \tag{16}$$

3.4 Quantifying Representation-Level Modality Gap

We quantify representation-level modality gap by modeling each modality's embeddings as an effective low-dimensional subspace in the shared representation space. Under dimension collapse, imbalance is reflected by the angular mismatch between modality-specific subspaces, which motivates a principal-angle-based measure of cross-modal representation misalignment. This formulation is inspired by recent analyses of modality-gap subspace geometry and dimension collapse in cross-modal representations Yi et al. (2025); Zhang et al. (2024b).

Formally, assume we have two modality feature matrices $\boldsymbol{Z}_x \in \mathbb{R}^{N\times d}$ and $\boldsymbol{Z}_y \in \mathbb{R}^{N\times d}$, where N is the number of paired samples and d is the embedding dimension. These modalities effectively reside in two low-dimensional subspaces $\mathcal{S}_x, \mathcal{S}_y \subseteq \mathbb{R}^d$. To extract the orthonormal bases of these respective subspaces, we perform Singular Value Decomposition (SVD):

$$\boldsymbol{Z}_x = \boldsymbol{P}_x\boldsymbol{\Sigma}_x\boldsymbol{Q}_x^\top, \qquad \boldsymbol{Z}_y = \boldsymbol{P}_y\boldsymbol{\Sigma}_y\boldsymbol{Q}_y^\top. \tag{17}$$

Crucially, because the subspaces are defined in the feature embedding space $\mathbb{R}^d$ (rather than the sample-index space), the basis matrices must be rigorously extracted from the right singular vectors:

$$\boldsymbol{U}_x = \boldsymbol{Q}_x[:, 1:k_x], \qquad \boldsymbol{U}_y = \boldsymbol{Q}_y[:, 1:k_y]. \tag{18}$$

This operation is analytically equivalent to computing the top eigenvectors of the covariance matrices $\boldsymbol{Z}_x^\top\boldsymbol{Z}_x$ and $\boldsymbol{Z}_y^\top\boldsymbol{Z}_y$. The retained subspace dimensions k_x and k_y can be dynamically selected by preserving a target explained variance ratio τ (e.g., 0.95):

$$\frac{\sum_{i=1}^{k}\lambda_i}{\sum_{i=1}^{d}\lambda_i} \geq \tau, \tag{19}$$

where λ_i denotes the eigenvalues of the corresponding covariance matrix.

Ultimately, we quantify the modality gap by measuring the smallest principal angle between $\mathcal{S}_x$ and $\mathcal{S}_y$. Given the orthonormal bases $\boldsymbol{U}_x$ and $\boldsymbol{U}_y$, we compute the overlap matrix $\boldsymbol{M} = \boldsymbol{U}_x^\top\boldsymbol{U}_y$. Let the singular values of $\boldsymbol{M}$ be ordered as $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k$, where $k = \min(k_x, k_y)$. The principal angles are defined as $\theta_i = \arccos(\sigma_i)$. We use the smallest principal angle to quantify the representation-level modality gap:

$$\theta_{\min} = \arccos\big(\min(1, \max(-1, \sigma_{\max}(\boldsymbol{U}_x^\top\boldsymbol{U}_y)))\big), \tag{20}$$

where the maximum singular value is strictly clipped into $[-1, 1]$ for numerical stability. Under this formulation, smaller angles indicate stronger subspace alignment, whereas larger angles reflect greater representation-level mismatch between modalities.

4 Experiments

4.1 Experimental Setup

Datasets. We evaluate CQFA on three widely used audio-visual classification benchmarks: CREMA-D, AVE, and Kinetics-400. CREMA-D Cao et al. (2014) consists of 7,442 audio-visual clips for emotion recognition across six classes. AVE Tian et al. (2018) contains 4,143 clips, each 10 seconds

long, annotated for audio-visual event classification across 28 classes. Kinetics-400 Kay et al. (2017) contains approximately 300,000 clips, each 10 seconds long, covering 400 human action classes.

Baselines. To evaluate CQFA under a unified modality-imbalance setting, we compare it with fourteen representative baselines: AUG Jiang et al. (2025) , OGM Peng et al. (2022), AGM Li et al. (2023), PMR Fan et al. (2023), MMPareto Wei & Hu (2024), LFM Yang et al. (2024), MLA Zhang et al. (2024a), TCMax Yu et al. (2026), MMCosine Xu et al. (2023), DIMML Fan et al. (2024), Remix Ma et al. (2025), QMF Zhang et al. (2023), DGL Wei et al. (2025a), and InfoReg Huang et al. (2025). These methods cover gradient modulation, objective balancing, representation alignment, and fusion-control strategies. For a fair comparison, all methods use the same pretrained CLIP visual encoder and CLAP audio encoder, and both encoders are fine-tuned with LoRA under identical configurations; therefore, the comparison mainly reflects differences in fusion and imbalance-mitigation mechanisms.

Implementation Details and Metrics. We use CLIP ViT-B/32 and CLAP HTS-AT as the visual and audio backbones, respectively, and adapt both encoders with LoRA. The visual input consists of 16 RGB frames resized to 224×224, while the audio input is resampled to 48 kHz and represented by 64-bin log-mel features. CQFA projects both modalities into a 512-dimensional query space and is trained on an NVIDIA RTX 4090 GPU. Detailed implementation settings are provided in Appendix A. We report Accuracy (ACC) and F1 score for classification performance.

4.2 Results on Different Benchmarks

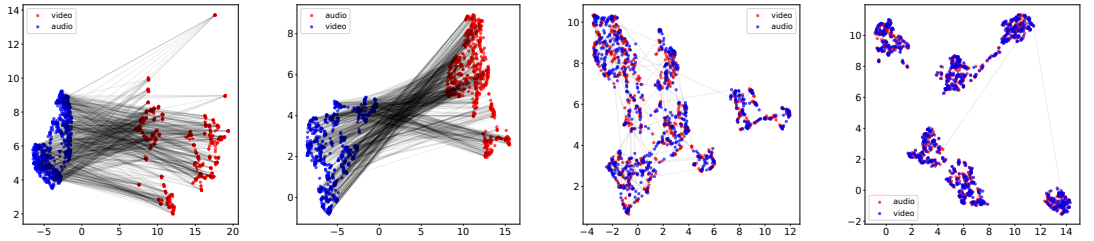
Table 1 reports the ACC and F1 scores of CQFA and competing methods on CREMA-D, AVE, and Kinetics-400. These three benchmarks cover audio-visual emotion recognition, event classification, and action recognition, allowing us to evaluate classification performance across different data scales and category granularities.

Table 1: Comparison of different methods on CREMA-D, AVE, and Kinetics-400. Metrics reported are Accuracy (ACC) and F1 Score.

Method	**Venue (Year)**	**CREMA-D**		**AVE**		**Kinetics-400**	
		ACC (%)	F1 (%)	ACC (%)	F1 (%)	ACC (%)	F1 (%)
OGM	CVPR 2022	85.46	85.68	96.52	96.83	69.87	69.52
AGM	ICCV 2023	86.04	85.93	95.02	95.10	65.66	64.96
PMR	CVPR 2023	86.49	86.44	95.02	94.95	70.18	69.69
MMCosine	ICASSP 2023	88.02	88.20	**97.76**	97.91	69.52	68.86
QMF	ICML 2023	87.39	87.44	96.27	96.25	70.14	69.66
MMPareto	ICML 2024	88.29	88.32	96.77	96.87	70.68	70.23
LFM	NeurIPS 2024	88.96	89.09	97.26	97.29	70.95	70.53
MLA	CVPR 2024	86.81	87.21	95.27	95.73	64.92	63.75
DIMML	ACM MM 2024	87.84	87.90	96.77	96.58	66.31	65.74
AUG	NeurIPS 2025	88.74	88.79	**97.76**	**97.93**	66.44	65.76
Remix	ICML 2025	87.75	88.01	96.77	97.00	70.50	70.06
DGL	ICCV 2025	87.09	87.13	97.01	97.08	70.13	69.70
InfoReg	CVPR 2025	86.14	86.56	95.52	95.52	65.36	64.39
TCMax	ICLR 2026	86.81	87.03	96.77	96.70	68.25	67.75
CQFA (Ours)		**89.94**	**89.94**	95.52	95.87	**71.85**	**71.66**

The results reveal two main observations. First, CQFA achieves state-of-the-art performance on CREMA-D and Kinetics-400, reaching 89.94/89.94 and 71.85/71.66 ACC/F1, respectively, but does not obtain the best result on AVE. A plausible explanation is that AVE is relatively small, and combining parameter-rich pretrained backbones with an additional Q-Former may increase the risk of overfitting. The query-based fusion architecture is therefore more suitable for larger and more diverse datasets, where its additional capacity can be more effectively exploited.

Second, LFM is the strongest non-CQFA baseline on CREMA-D and Kinetics-400, which is consistent with its use of contrastive learning to reduce cross-modal discrepancy. However, the effective-



(a) Backbone (v/a) (b) Specialist queries (q_a/q_v) (c) AVC embeddings (g_v/g_a) (d) Fusion streams (q_f^a/q_f^v)

Figure 2: Layer-wise UMAP gap visualizations of CQFA on the CREMA-D dataset.

dimension analysis in Table 4 shows that LFM still exhibits a large audio-video effective-dimension gap, indicating that it does not fully resolve effective-dimensional collapse. Thus, although LFM narrows the modality gap, its representational capacity is not fully released, which limits further performance improvement.

4.3 Ablation Studies

We evaluate training objectives, architectural components, and AVM-weight sensitivity in separate experiments. The objective ablation in Appendix C shows that removing AVC lowers performance, AVM has dataset-dependent effects, and removing CLS yields near-random results. For the architectural ablation, we retain all objectives and vary one structural component at a time; Table 2 reports ACC/F1 from one validation-selected checkpoint per configuration.

Table 2: Architectural ablation of CQFA (ACC/F1).

Variant	CREMA-D	AVE	Kinetics-400
CQFA	**89.94/89.94**	94.03/94.35	**71.85/71.66**
w/o Bottleneck Mask	87.84/87.88	93.28/94.10	69.68/69.15
Audio-led Stream	88.44/88.46	94.28/94.74	70.27/69.90
Video-led Stream	88.14/88.11	**95.52/95.90**	70.43/70.10
Feature Fusion	87.99/87.93	94.03/94.48	70.11/69.83
w/o Query Partition	86.64/86.64	94.78/95.33	69.56/69.30
Capacity-Matched Sharing	87.99/87.99	93.78/94.39	69.36/69.08
w/o Text Prototypes	88.89/88.92	93.53/94.01	70.53/70.19

Table 2 shows that full CQFA outperforms all architectural controls on CREMA-D and Kinetics-400, with ACC margins of 1.05–3.30 and 1.32–2.49 points, respectively. The largest losses arise from removing query partitioning on CREMA-D and replacing structured routing on Kinetics-400. Some controls perform better on AVE, indicating dataset-dependent sensitivity. Exact parameter parity of the capacity-matched sharing control remains unverified. With fusion queries fixed, logit-level fusion improves ACC on CREMA-D and Kinetics-400 but not AVE (Appendix D). Permutation tests show that CQFA uses sample-specific audio information, although reliance on video is greater on CREMA-D and Kinetics-400 (Appendix E). On AVE, $\alpha_{\text{AVM}} = 0$ yields higher ACC and F1 than 0.20 at the evaluated seed (Appendix F).

4.4 Modality Gap Visualization

We visualize the layer-wise evolution of CQFA's modality gap using pairwise UMAP projections of normalized audio and visual embeddings, as shown in Figure 2. The complete method-level visualizations are provided in Appendix K. We further quantify the modality gap with complementary distance-based metrics in Table 3, with the complete quantitative results reported in Appendix H.

Since the method-level statistics are computed from backbone embeddings, they mainly reflect the modality gap before high-level alignment and fusion. Table 3 shows that CQFA does not achieve the smallest backbone-level gap; for example, AUG has lower backbone-stage L2 and cosine distances,

Table 3: Selected modality-gap statistics on the CREMA-D dataset.

Method	L2	Cosine	UMAP Distance	UMAP Center
MMCosine	1.417483	1.005165	14.791110	14.683060
LFM	1.409830	0.994628	20.706594	20.601686
AUG	1.392287	0.969682	17.344053	17.266756
CQFA (Ours)	1.419171	1.007265	17.820297	17.378708
Backbone (v/a)	1.420660	1.009395	18.392397	17.875828
Specialist queries (q_a/q_v)	1.476505	1.092080	17.096130	16.977249
AVC embeddings (g_v/g_a)	0.711697	0.263736	0.527949	0.041841
Fusion streams (q_f^a/q_f^v)	0.289861	0.046931	0.115583	0.020094

yet it does not achieve state-of-the-art classification performance. This suggests that reducing the raw embedding-level gap alone is insufficient to fully alleviate representation-level imbalance. In CQFA, the substantial gap reduction emerges after the AVC embeddings (g_v/g_a), where both the quantitative metrics and UMAP visualization become much more compact, indicating that task-aware contrastive alignment rather than backbone-level gap minimization is the key mechanism for releasing balanced multimodal representations.

4.5 Effective Dimension Analysis

Following Eq. (1), we compute the effective dimension d_e as the minimum number of principal directions whose cumulative singular-value energy explains 99% of the representation variance. Thus, d_e measures how many dominant directions are needed to explain most of the feature space, while the inter-modal gap $|\Delta_d| = |d_e^v - d_e^a|$ reflects representation-level imbalance between audio and visual subspaces. Table 4 reports the effective dimensions of different modalities and their gap for selected methods, with the complete results and cumulative explained-variance curves provided in Appendix I and Appendix J, respectively.

Table 4: Selected Effective-Dimension Results.

Method	$d_e^a \uparrow$	d_e^v	$\lvert\Delta_d\rvert$
MMCosine	**217**	**236**	19
LFM	157	203	46
AUG	159	198	39
CQFA (Ours)	165	177	12

As shown in Table 4, MMCosine obtains the highest effective dimensions for both modalities, which is consistent with its strong classification performance among the baselines. CQFA does not achieve the largest effective dimensions, nor does it obtain the smallest dimension gap in the complete comparison. Nevertheless, its effective dimensions and $|\Delta_d|$ remain in a moderate range compared with other methods, suggesting that CQFA alleviates representation-level imbalance to a certain extent rather than relying on maximal backbone dimensionality. Additional t-SNE visualizations are provided in Appendix L.

5 Conclusion

In this work, we studied modality imbalance as both representation-level subspace mismatch and fusion-level weak-modality underuse. We proposed CQFA, a query-guided framework that uses staged alignment and decoupled logits fusion to improve audio-visual learning. Experiments on CREMA-D and Kinetics-400 show that CQFA achieves strong performance, and the geometric analyses suggest that its gains are more closely tied to high-level cross-modal alignment and controlled fusion than to raw backbone-gap reduction alone.

Limitation. Although CQFA shows strong results on the evaluated benchmarks, our experiments do not yet include broader comparisons across more audio-visual datasets. Future work will evaluate CQFA on a wider range of datasets to further verify its generalization ability under different data distributions and task settings.

AI USE STATEMENT

Generative AI tools were used to assist with method implementation, development of portions of the code, and analysis of experimental results, as well as manuscript editing, LaTeX formatting, and reference checking. The authors take responsibility for all AI-assisted content and the final manuscript.

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A Implementation Details

Table 5: Detailed implementation settings for CQFA.

Item	Setting
Visual backbone	CLIP ViT-B/32 (`openai/clip-vit-base-patch32`); vision-token hidden size 768 and pooled projection size 512.
Audio backbone	CLAP HTS-AT (`laion/clap-htsat-unfused`); audio-token hidden size 768 and pooled projection size 512.
Visual input	16 RGB frames per video, with each frame resized to $3 \times 224 \times 224$.
Audio input	48 kHz waveform, cropped or padded to 10 s and converted into 64-bin log-mel features.
Model type	`dlip_bf3_logitsfusion`.
Shared representation	Visual, audio, and text features are projected to hidden dimension $D = 512$.
LoRA adaptation	Rank 4, alpha 16, dropout 0.01, and no bias; CLIP uses `q_proj`/`v_proj`, CLAP uses `query`/`value`, and text encoders are frozen.
Q-Former	4 layers, 8 attention heads, 8 visual expert queries, 8 audio expert queries, and 4 fusion queries.
Logits fusion	Two fusion streams are used, and the final prediction sums the CLIP-space and CLAP-space similarity logits.
Training objectives	$\alpha_{\text{AVC}} = 1.0$, $\alpha_{\text{AVM}} = 0.5$, and $\alpha_{\text{CLS}} = 1.0$.
Contrastive temperature	0.07.
ITM negative sampling	Hard negative sampling.
Optimization	Adam optimizer with learning rate 1×10^{-4} and StepLR scheduler with step size 10 and decay factor 0.1.
Training schedule	30 epochs, batch size 32, and random seed 42.
Hardware	NVIDIA RTX 4090 GPU.
Evaluation metrics	Accuracy (ACC) and F1 score.

B Three-Stage Training Details

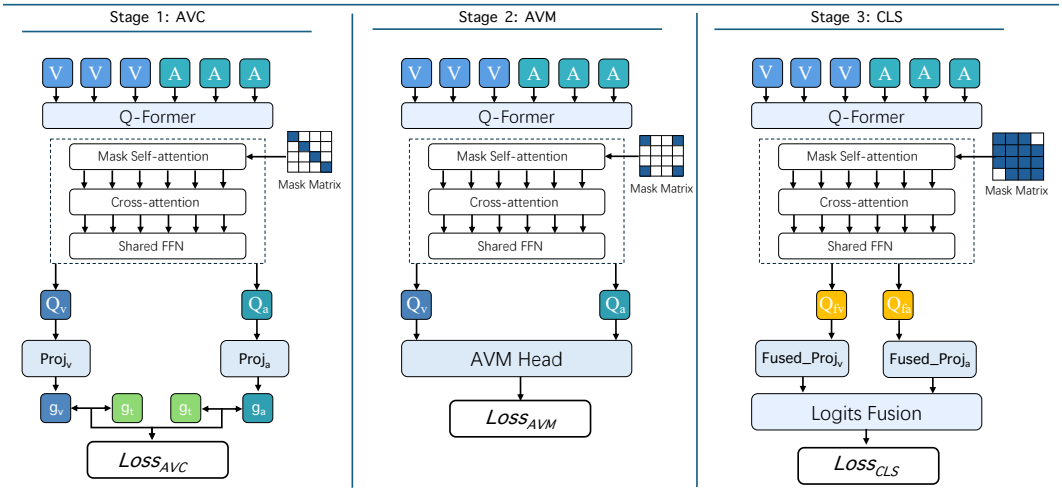


Figure 3: Three-stage training in CQFA.

C Objective Contributions

We evaluate the contribution of each training objective while holding the CQFA architecture fixed. This experiment isolates objective-level effects and is therefore conceptually distinct from the architectural comparisons in Section 4.3.

Table 6: Objective ablation of CQFA on the three evaluation benchmarks.

Variant	CREMA-D		AVE		Kinetics-400	
	ACC	F1	ACC	F1	ACC	F1
Full CQFA	**89.94**	**89.94**	95.52	95.87	71.85	**71.66**
w/o AVC	85.74	85.67	95.27	95.54	71.25	70.97
w/o AVM	89.34	89.30	**96.02**	**96.26**	**71.88**	71.62
w/o CLS	18.77	15.30	1.99	1.40	0.33	0.04

As reported in Table 6, removing AVC consistently reduces both ACC and F1, with the largest degradation observed on CREMA-D. This result supports the contribution of contrastive and prototype-based supervision to modality-specific representation learning. The effect of AVM is less uniform: its removal decreases performance on CREMA-D, improves both metrics on AVE, and produces a marginal ACC increase but a slight F1 decrease on Kinetics-400. AVM therefore provides dataset-dependent rather than uniformly positive benefits. The near-random performance of `w/o CLS` establishes only that AVC and AVM cannot substitute for task-level supervision. It does not provide evidence for query-role partitioning, directional fusion streams, or logit-level aggregation; these components are evaluated using classification-preserving controls in Section 4.3.

D Query Source and Fusion Level

We further disentangle the classification-query source from the aggregation level through the 2×2 factorial comparison in Table 7.

Table 7: Factorial comparison of classification-query source and fusion level. Results are reported as ACC/F1.

Query	Fusion level	CREMA-D	AVE	Kinetics-400
Fusion	Logits	**89.94/89.94**	**94.03/94.35**	**71.85/71.66**
Fusion	Features	87.99/87.93	**94.03/94.48**	70.11/69.83
Expert	Logits	88.14/88.11	93.53/93.79	69.30/69.03
Expert	Features	87.99/87.94	93.78/94.35	69.16/69.03

Holding the query source fixed to fusion queries, logit-level aggregation improves ACC over feature-level aggregation by 1.95 points on CREMA-D and 1.74 points on Kinetics-400, while producing identical ACC on AVE. Conversely, under logit-level aggregation, fusion queries outperform expert queries by 1.80, 0.50, and 2.55 ACC points on CREMA-D, AVE, and Kinetics-400, respectively. The magnitude of each effect consequently depends on the setting of the other factor, indicating an interaction between query source and fusion level rather than two universally additive contributions.

E Unimodal Capability and Decision-Level Utilization

Representation-space geometry alone is insufficient to establish whether the final classifier uses sample-specific information from the weaker modality. We therefore first estimate unimodal capability using independently trained audio-only and video-only models (Table 8). Audio is the weaker modality on all three benchmarks, with audio–video ACC gaps of 14.27, 1.49, and 41.20 points on CREMA-D, AVE, and Kinetics-400, respectively. Nevertheless, the audio-only models remain substantially above chance, confirming that the weaker modality retains discriminative information.

Table 8: Performance of independently trained unimodal models and the controlled CQFA baseline. Results are reported as ACC/F1.

Dataset	Audio-only	Video-only	CQFA
CREMA-D	63.06/62.52	77.33/77.19	89.94/89.94
AVE	85.57/83.06	87.06/87.72	94.03/94.35
Kinetics-400	23.64/22.20	64.84/64.51	71.85/71.66

To assess whether CQFA exploits this information, we keep the trained checkpoint fixed and independently permute either audio or video across samples. For each modality, we use five fixed-point-free dataset-level permutations and perform no retraining. We quantify the resulting decision-level reliance as

$$C_a = \mathrm{ACC}_{\mathrm{paired}} - \mathbb{E}_\pi[\mathrm{ACC}(a_\pi, v)], \qquad C_v = \mathrm{ACC}_{\mathrm{paired}} - \mathbb{E}_\pi[\mathrm{ACC}(a, v_\pi)], \tag{21}$$

where π denotes a fixed-point-free permutation. Larger values indicate greater sensitivity to disrupting the sample-wise correspondence of the associated modality.

Table 9: Fixed-checkpoint modality-permutation analysis. Shuffled results are reported as mean±standard deviation over five fixed-point-free permutations.

Dataset	Paired ACC	Audio shuffled	Video shuffled	C_a	C_v
CREMA-D	89.94	63.54 ± 1.87	31.02 ± 2.60	26.40	58.92
AVE	94.03	41.59 ± 2.03	46.97 ± 1.24	52.44	47.06
Kinetics-400	71.85	58.67 ± 0.19	1.73 ± 0.06	13.18	70.12

Audio permutation reduces ACC on every dataset, providing direct evidence that CQFA uses sample-specific information from the weaker modality. The substantially larger C_v values on CREMA-D and Kinetics-400 indicate stronger reliance on video correspondence, particularly on Kinetics-400, whereas the similar C_a and C_v values on AVE indicate a more balanced sensitivity to the two modalities. These results characterize prediction-level reliance under a distribution-shift intervention; they neither constitute a causal decomposition of modality contributions under the original data distribution nor directly measure interference among fusion neurons.

F AVM-Weight Sensitivity on AVE

Finally, we examine whether the AVE results are sensitive to the AVM objective weight. We vary only α_{AVM} while holding the architecture, optimization protocol, $\alpha_{\mathrm{AVC}} = 1.0$, and $\alpha_{\mathrm{CLS}} = 1.8$ fixed; each configuration is trained independently.

Table 10: Sensitivity to the AVM objective weight on AVE. Each configuration is trained independently.

α_{AVM}	Best epoch	ACC	F1	mAP
0.00	4	**96.02**	**96.24**	**99.00**
0.10	8	95.52	95.91	98.33
0.20 (default)	13	95.27	95.71	97.87
0.25	6	95.27	95.58	98.62
0.50	4	95.27	95.45	98.97

At the evaluated seed, setting $\alpha_{\mathrm{AVM}} = 0$ improves ACC and F1 by 0.75 and 0.53 points, respectively, relative to the default weight of 0.20; none of the nonzero settings attains comparable performance. Together with the video-led single-stream result in Table 2, this finding indicates that AVE is sensitive to both objective weighting and routing configuration. Because the sweep is based on a single seed and validation-set measurements, it does not establish statistically persistent negative transfer or identify a unique explanation for the observed behavior.

G PRINCIPAL-ANGLE-BASED MODALITY GAP ANALYSIS

This section measures the angular discrepancy between audio and visual representation subspaces as a structural indicator of modality gap. Table 11 reports the minimum nonzero principal angle for different methods and for layer-wise CQFA representations.

Table 11: Minimum nonzero principal-angle statistics on the CREMA-D dataset.

Entry	$\theta_{\min}^{nz}$ (rad) ↓	$\theta_{\min}^{nz}$ (deg) ↓
OGM	0.349	19.97
AGM	0.412	23.63
PMR	0.374	21.42
MMCosine	**0.104**	**5.98**
QMF	0.313	17.93
MMPareto	0.303	17.33
LFM	0.283	16.21
MLA	0.286	16.41
DIMML	0.242	13.89
AUG	0.325	18.64
Remix	0.259	14.81
DGL	0.389	22.31
InfoReg	0.398	22.80
TCMax	0.252	14.46
CQFA (Ours)	0.347	19.89
Backbone (v/a)	0.349	20.01
Projected tokens (v/a)	0.355	20.32
Specialist queries (q_a/q_v)	0.414	23.72
AVC embeddings (g_v/g_a)	0.083	4.77
Fusion streams (qf_a/qf_v)	0.045	2.60

The results show that CQFA does not reduce the subspace angle at the raw embedding stage: its backbone-level angle remains larger than several baselines, indicating that the initial backbone representations are not the main source of its final gain. However, the layer-wise CQFA results show a clear reduction after the AVC embeddings (g_v/g_a), where the angle decreases to 4.77°. This indicates that CQFA reduces the modality gap primarily through the subsequent task-aware AVC alignment stage rather than through backbone-level embedding alignment alone.

H Complete Modality-Gap Statistics

Table 12: Complete modality-gap statistics on the CREMA-D dataset.

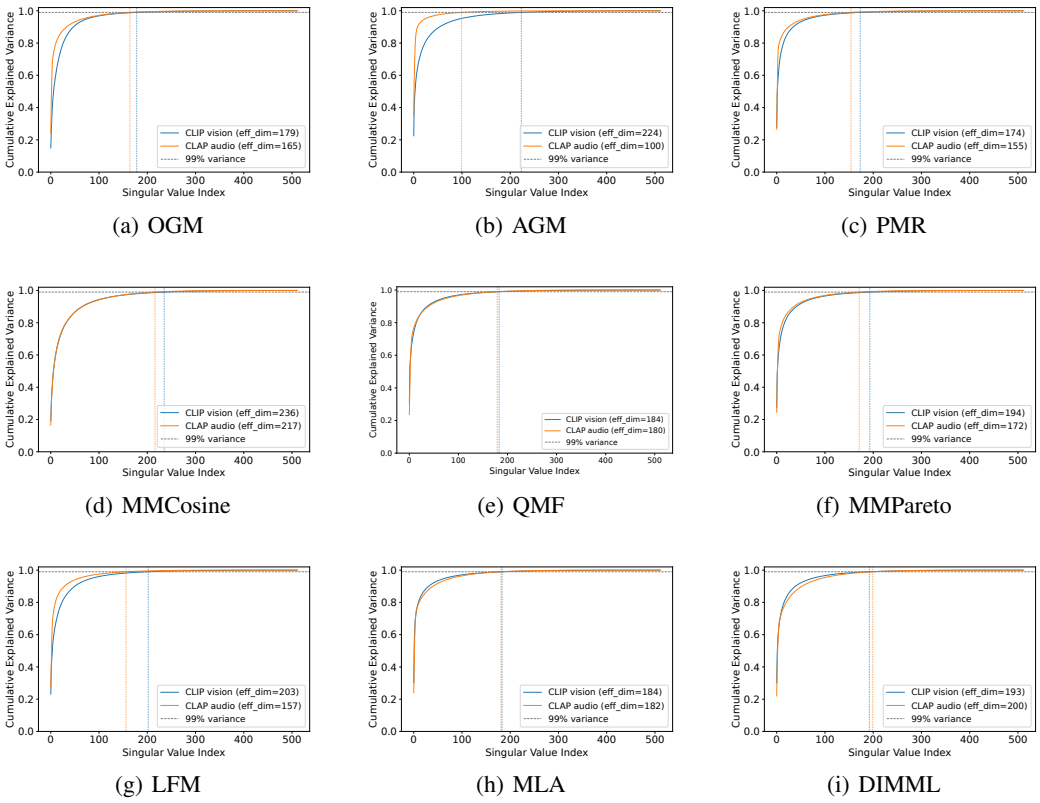
Method	L2	Cosine	UMAP Distance	UMAP Center
OGM	1.408391	0.992246	19.046309	18.885324
AGM	1.417963	1.005683	14.388806	14.137403
PMR	1.406326	0.989707	17.362234	17.189222
MMCosine	1.417483	1.005165	14.791110	14.683060
QMF	1.412988	0.999114	16.017960	15.868993
MMPareto	1.417956	1.005728	18.995197	18.857359
LFM	1.409830	0.994628	20.706594	20.601686
MLA	1.420102	1.009273	17.066839	16.981440
DIMML	1.417675	1.005854	16.741369	16.552137
AUG	1.392287	0.969682	17.344053	17.266756
Remix	1.415861	1.002840	17.958792	17.876984
DGL	1.405201	0.988407	16.339041	16.248043
InfoReg	1.410580	0.995524	18.193207	18.083967
TCMax	1.414488	1.001186	21.979671	21.924444
CQFA (Ours)	1.419171	1.007265	17.820297	17.378708
Backbone (v/a)	1.420660	1.009395	18.392397	17.875828
Specialist queries (q_a/q_v)	1.476505	1.092080	17.096130	16.977249
AVC embeddings (g_v/g_a)	0.711697	0.263736	0.527949	0.041841
Fusion streams (q_f^a/q_f^v)	0.289861	0.046931	0.115583	0.020094

I Complete Effective-Dimension Statistics

Table 13: Complete effective dimensions measured at 99% variance for audio and video backbone representations.

Method	$d_e^a \uparrow$	d_e^v	$\lvert\Delta_d\rvert$
OGM	165	179	14
AGM	100	224	124
PMR	155	174	19
MMCosine	**217**	**236**	19
QMF	180	184	4
MMPareto	172	194	22
LFM	157	203	46
MLA	182	184	2
DIMML	200	193	7
AUG	159	198	39
Remix	193	193	**0**
DGL	165	156	9
InfoReg	167	154	13
TCMax	178	207	29
CQFA (Ours)	165	177	12

J Complete Cumulative Explained-Variance Curves

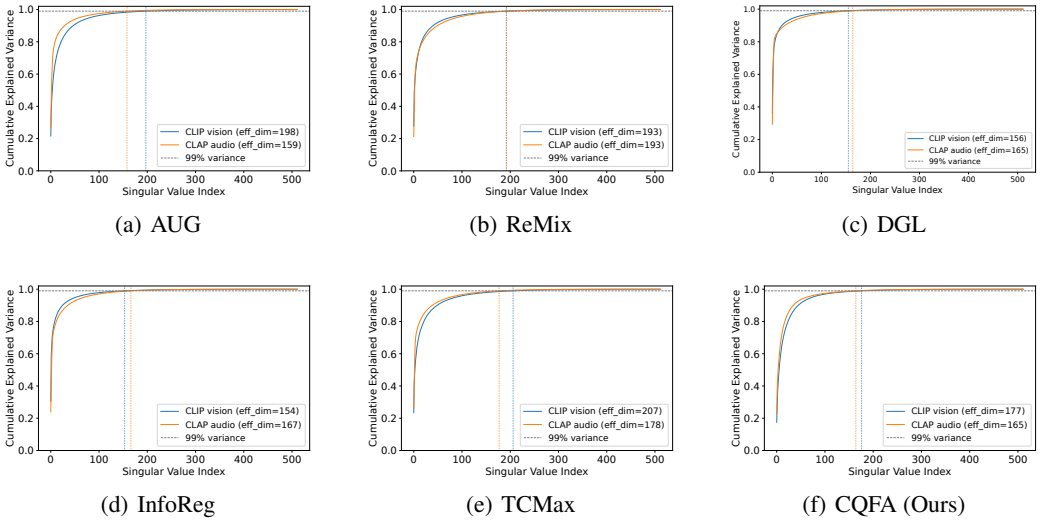


(a) OGM (b) AGM (c) PMR

(d) MMCosine (e) QMF (f) MMPareto

(g) LFM (h) MLA (i) DIMML

Figure 4: Complete cumulative explained-variance curves of audio and visual backbone representations, Part I.



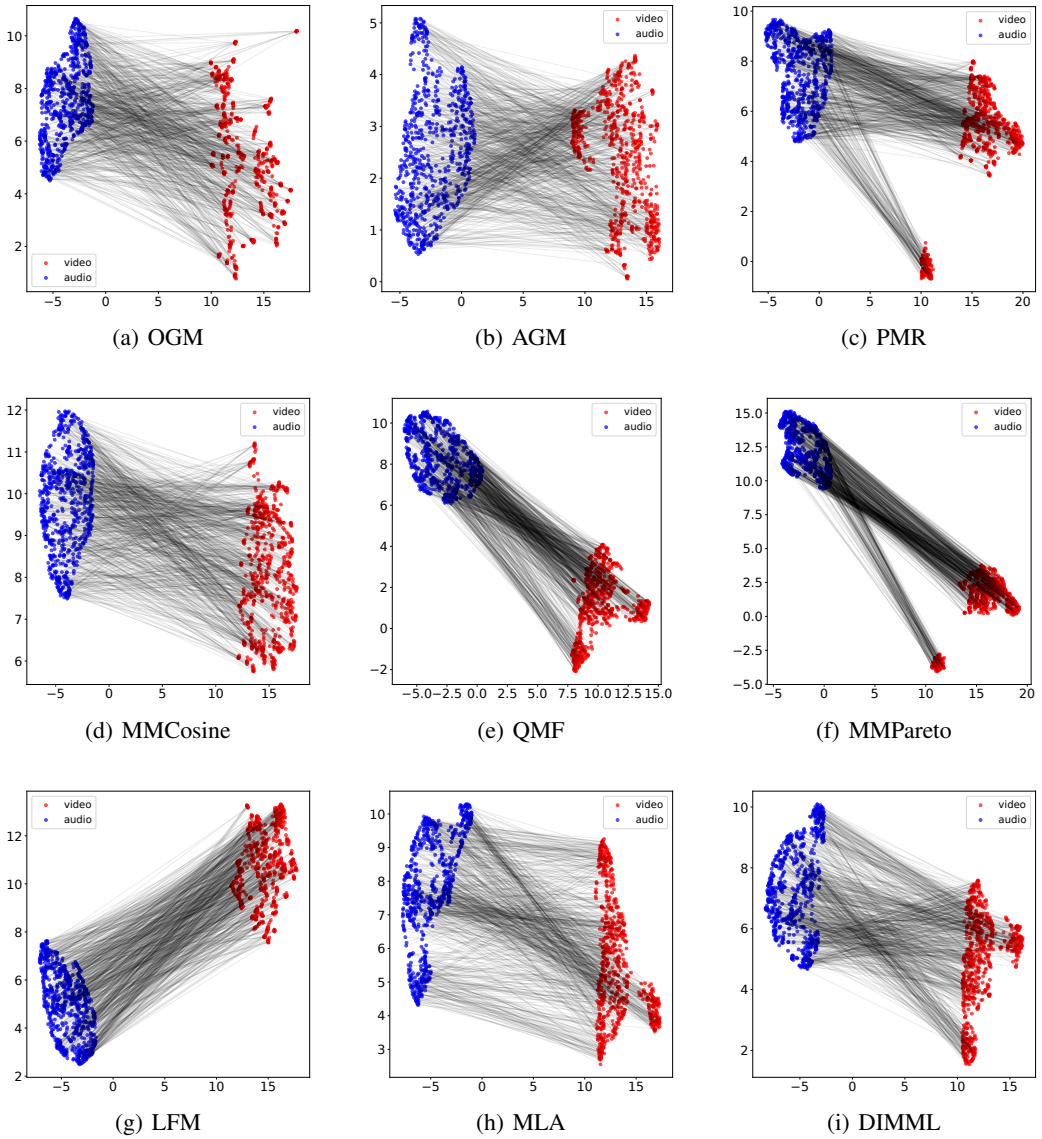
(a) AUG (b) ReMix (c) DGL

(d) InfoReg (e) TCMax (f) CQFA (Ours)

Figure 5: Complete cumulative explained-variance curves of audio and visual backbone representations, Part II.

K Additional Modality Gap Visualizations

For completeness, we provide the full set of pairwise UMAP visualizations. The methods are ordered consistently with Table 1 whenever applicable.

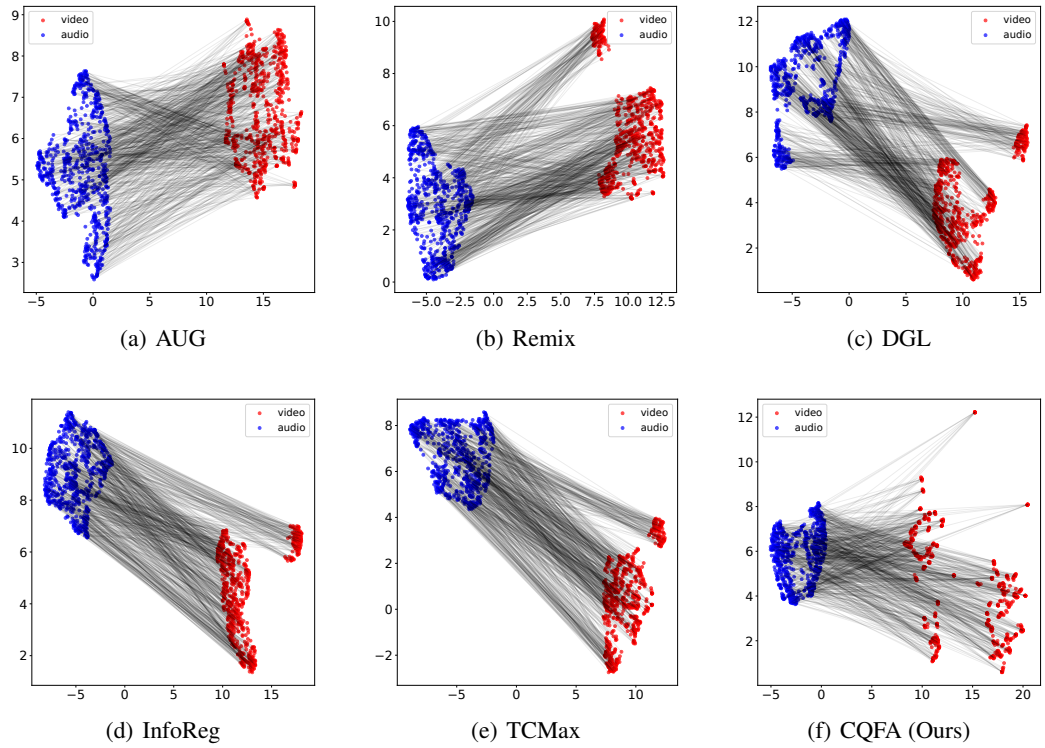


(a) OGM (b) AGM (c) PMR

(d) MMCosine (e) QMF (f) MMPareto

(g) LFM (h) MLA (i) DIMML

Figure 6: Complete pairwise UMAP visualizations of normalized backbone embeddings on the CREMA-D dataset, Part I. Vision features are shown in red, Audio features in blue, and gray lines connect paired samples.



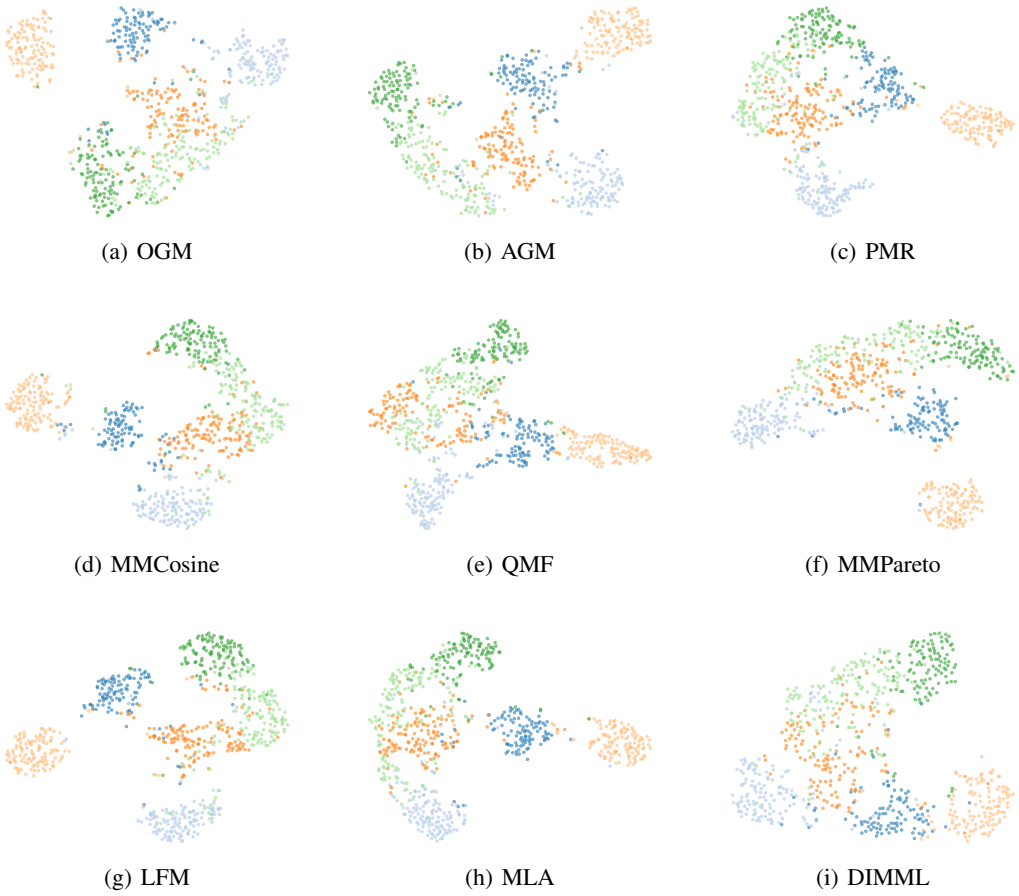
(a) AUG (b) Remix (c) DGL

(d) InfoReg (e) TCMax (f) CQFA (Ours)

Figure 7: Complete pairwise UMAP visualizations of normalized backbone embeddings on the CREMA-D test set, Part II.

L ADDITIONAL T-SNE VISUALIZATIONS

For completeness, we provide the full set of t-SNE visualizations. The methods are ordered consistently with Table 1 whenever applicable.



(a) OGM (b) AGM (c) PMR

(d) MMCosine (e) QMF (f) MMPareto

(g) LFM (h) MLA (i) DIMML

Figure 8: Complete t-SNE visualizations of normalized backbone embeddings on the CREMA-D test set, Part I.



(a) AUG (b) ReMix (c) DGL



(d) InfoReg (e) TCMax (f) CQFA (Ours)

Figure 9: Complete t-SNE visualizations of normalized backbone embeddings on the CREMA-D test set, Part II.